Project Design Phase | Proposed Solution Template

Date: 27-10-2026 | **Team ID:** SWUID2026403175 | **Project Name:** Agri-Tech | **Maximum Marks:** 2 Marks

Proposed Solution for Agri-Tech

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Many farmers struggle to make timely decisions because they lack accurate weather forecasts, soil insights, early pest alerts and easy access to experts and resources. Limited digital tools in local languages lead to crop loss, lower yield and uncertainty.
2	Idea / Solution Description	Agri-Tech is a full-stack MERN web application (React frontend + Node.js/Express backend + MongoDB) that lets farmers manage farm profiles and crops, view weather forecasts, receive pest alerts, analyse soil health, order resources and consult experts. Admins manage users, content and resource approvals.
3	Novelty / Uniqueness	- Single platform combining weather, soil, pest, resources and experts - Responsive, multilingual interface for varied literacy levels - Role-based security with JWT and bcrypt - Interactive charts and graphical reports for decisions - Community forums for peer learning
4	Social Impact / Customer Satisfaction	- Helps farmers increase yield and reduce crop loss - Promotes sustainable and data-driven agriculture - Bridges the gap between traditional and modern farming - Improves access to expert knowledge in rural regions
5	Business Model (Revenue Model)	- Free basic access for farmers - Commission on resource sales through the marketplace - Premium subscription for advanced analytics and priority expert consultation - Partnerships with agri-input suppliers and agri-experts
6	Scalability of the Solution	Expandable to new regions with additional languages and crop data. Modular architecture allows integration of IoT sensors, mobile apps (React Native), AI-based crop recommendations and payment gateways. Optimized MongoDB queries and caching support large datasets.